

Investigation Report: suspicious_activity.log

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Summary:

The suspicious_activity.log file reveals strong indicators of malicious post-exploitation behaviour. The log shows PowerShell commands being executed using obfuscation and system manipulation techniques commonly associated with malware activity, particular with the intent to evade Windows Defender.

Key Findings:

- Suspicious PowerShell Execution:

Example Command:

```
powershell.exe -nop -w hidden -EncodedCommand ...
```

- This technique used EncodedCommand to obfuscate malicious payloads. The -nop (no profile) and -w hidden (hidden window) flags are often used by threat actors to execute PowerShell silently and avoid detection.
- **Use of Add-MpPreference:**
 - This command modifies Windows Defender preferences to exclude specific paths, processes from being scanned, or file types.
 - It is often abused in post-exploitation stages to evade detection of malicious files or tools.

- Detection was captured by the Sigma rule: rule1_add-mppreference.yaml

- **Execution Context:**
 - The user executing the commands appears to have SYSTEM or Administrator privileges.
 - This elevated context means the attacker already achieved privilege escalation or has administrative access, which is a red flag in incident response.

- **Event Correlation:**
 - The timestamp aligns with other potential malicious artifacts.
 - Execution behaviours resemble known TTPs from threat actors using fileless malware, evasion frameworks, and living off the land binaries (LOLBins).

Conclusion:

There is clear evidence of malicious PowerShell use designed to bypass security controls, maintain persistence, and potentially exfiltrate data or establish C2 (Command & Control). This log should be treated as part of an active or recent compromise.

Recommendations:

1. Isolate the affected host from the network immediately.
2. Hunt for similar behaviour on other systems, especially encoded PowerShell executions and Add-MpPreference abuse.
3. Review EDR logs and process trees for suspicious command-line activity.

4. Enforce stricter PowerShell policies, disable PowerShell v2, and audit script block logging
5. Collect memory dumps and volatile data for forensic analysis.
6. Consider threat hunting for follow-up actions like lateral movement, C2 connections, and credential theft.